30-Day SQL Course

Target: Data Engineer / Azure Data Engineer / Microsoft Fabric Data Engineer
Daily time: 2–3 hours
Practice: 100 SQL problems
Database: SQL Server / Azure SQL syntax
Final goal: Be able to solve real-world SQL interview problems without relying on memorized queries.

Datasets You Will Use

Use one main business dataset throughout the course so that the problems become progressively harder.

Dataset 1 — Customers

Column	Type
CustomerID	INT
CustomerName	VARCHAR
Email	VARCHAR
City	VARCHAR
State	VARCHAR
Country	VARCHAR
RegistrationDate	DATE
CustomerSegment	VARCHAR

Dataset 2 — Products

Column	Type
ProductID	INT

ProductName	VARCHAR
Category	VARCHAR
SubCategory	VARCHAR
Price	DECIMAL
Cost	DECIMAL

Dataset 3 — Orders

Column	Type
OrderID	INT
CustomerID	INT
OrderDate	DATE
OrderStatus	VARCHAR
SalesAmount	DECIMAL
ShippingCost	DECIMAL

Dataset 4 — OrderDetails

Column	Type
OrderDetailID	INT
OrderID	INT
ProductID	INT
Quantity	INT
UnitPrice	DECIMAL
Discount	DECIMAL

Dataset 5 — Employees

Column	Type
EmployeeID	INT

EmployeeName	VARCHAR
DepartmentID	INT
ManagerID	INT
Salary	DECIMAL
HireDate	DATE

Dataset 6 — Departments

Column	Type
DepartmentID	INT
DepartmentName	VARCHAR
Location	VARCHAR

Dataset 7 — Transactions

Column	Type
TransactionID	INT
CustomerID	INT
TransactionDate	DATE
TransactionType	VARCHAR
Amount	DECIMAL
Status	VARCHAR

Dataset 8 — Healthcare Claims

This is useful for practicing realistic Data Engineering scenarios.

Column	Type
ClaimID	BIGINT
PatientID	INT

ProviderID	INT
ClaimDate	DATE
ClaimAmount	DECIMAL
PaidAmount	DECIMAL
ClaimStatus	VARCHAR
DiagnosisCode	VARCHAR

Week 1 — SQL Fundamentals

Day 1 — Database & SQL Fundamentals

Theory

Learn:

- Database
- DBMS
- RDBMS
- SQL
- Tables
- Rows
- Columns
- Schema
- Primary key
- Foreign key
- NULL
- Data types
- SQL Server basics

Practice

1. Create Customer table.
2. Create Product table.
3. Create Orders table.
4. Insert 10 customers.
5. Insert 10 products.
6. Insert 20 orders.
7. Display all customers.
8. Display all products.

9. Display customer names.
10. Display order IDs and amounts.

Interview Questions

- What is a database?
- DBMS vs RDBMS?
- What is a primary key?
- What is a foreign key?
- Can a primary key contain NULL?
- What is a schema?
- What is normalization?

Day 2 — SELECT, WHERE & Filtering

Theory

Learn:

SELECT
FROM
WHERE
AND
OR
NOT
IN
BETWEEN
LIKE
IS NULL

Practice

11. Find customers from USA.
12. Find customers from California.
13. Find products above $100.
14. Find products between $50 and $200.
15. Find orders above $1,000.
16. Find completed orders.
17. Find customers whose name starts with A.
18. Find products containing "Phone".
19. Find orders from 2026.
20. Find customers where email is NULL.

Interview

- Difference between `WHERE` and `HAVING`?
- How does `LIKE` work?
- How does NULL behave?
- Difference between `=` and `IS NULL`?

Day 3 — Sorting, DISTINCT & TOP

Theory

Learn:

ORDER BY
DISTINCT
TOP

Practice

21. Find unique countries.
22. Find unique customer segments.
23. Top 10 customers.
24. Top 5 expensive products.
25. Lowest 5 product prices.
26. Sort customers alphabetically.
27. Sort products by price descending.
28. Sort orders by date.
29. Find top 3 largest orders.
30. Find top 10 products by price.

Interview

- What does DISTINCT do?
- DISTINCT vs GROUP BY?
- TOP vs ROW_NUMBER?
- How do you find top N records?

Day 4 — Aggregate Functions

Theory

Learn:

COUNT
SUM
AVG
MIN
MAX

Practice

31. Count customers.
32. Count products.
33. Calculate total sales.
34. Calculate average order amount.
35. Find maximum order.
36. Find minimum order.
37. Calculate total quantity sold.
38. Find average product price.
39. Find total shipping cost.
40. Count completed orders.

Interview

- `COUNT(*)` vs `COUNT(column)`?
- How does COUNT handle NULL?
- SUM vs COUNT?
- Can AVG ignore NULL?

Day 5 — GROUP BY & HAVING

Theory

Understand the SQL execution concept:

FROM
WHERE
GROUP BY
HAVING
SELECT
ORDER BY

Practice

41. Sales by customer.
42. Sales by country.
43. Sales by product.

44. Sales by category.
45. Orders per customer.
46. Customers with more than 5 orders.
47. Products sold more than 100 units.
48. Average salary by department.
49. Departments with average salary > $80,000.
50. Customers spending more than $10,000.

Interview

- WHERE vs HAVING?
- Can HAVING be used without GROUP BY?
- Can GROUP BY contain multiple columns?
- What happens if you select a column not in GROUP BY?

Day 6 — SQL Joins

Theory

Master:

INNER JOIN
LEFT JOIN
RIGHT JOIN
FULL OUTER JOIN
CROSS JOIN
SELF JOIN

Practice

51. Customers with orders.
52. Customers without orders.
53. Orders with customer names.
54. Products with order details.
55. Customer + Order + Product.
56. Employees + Departments.
57. Employees with managers.
58. Customers who purchased products.
59. Products never purchased.
60. Departments with no employees.

Interview

- INNER JOIN vs LEFT JOIN?

- Why can a JOIN create duplicate rows?
- LEFT JOIN with WHERE condition?
- JOIN condition in ON vs WHERE?
- What is a Cartesian product?

Day 7 — Week 1 Project

Build:

Customer Sales Report

Produce:

CustomerID
CustomerName
Country
TotalOrders
TotalQuantity
TotalSales
AverageOrderValue
LastOrderDate

Requirements:

- Use joins.
- Use aggregation.
- Use GROUP BY.
- Use CASE.
- Handle NULL.
- Sort by total sales.

Week 1 Interview

You should be able to explain your query **line by line**.

Week 2 — Intermediate SQL

Day 8 — CASE Statements

Learn:

CASE
WHEN
THEN
ELSE
END

Practice:

- Salary categories
- Customer categories
- Order status classification
- Profit calculation
- Discount classification
- High-value customers
- Product profitability

Day 9 — NULL Handling

Learn:

ISNULL()
COALESCE()
NULLIF()
IS NULL
IS NOT NULL

Practice:

- Replace NULL with 0.
- Replace missing country.
- Handle NULL discounts.
- Avoid division by zero.
- Compare NULL values.
- Calculate sales with missing values.

Day 10 — String Functions

Learn:

UPPER
LOWER
TRIM

LEN
LEFT
RIGHT
SUBSTRING
CHARINDEX
REPLACE
CONCAT

Practice:

- Clean customer names.
- Extract email domain.
- Find names by pattern.
- Standardize country names.
- Extract phone prefixes.
- Remove unwanted characters.

Day 11 — Date Functions

Learn:

GETDATE()
DATEADD()
DATEDIFF()
DATEPART()
DATENAME()
EOMONTH()
YEAR()
MONTH()
DAY()

Practice:

- Orders by year.
- Orders by month.
- Monthly sales.
- Customers registered this year.
- Customer age.
- Days between orders.
- Month-end reporting.
- Previous month.
- Next month.
- Orders in last 30 days.

Day 12 — Subqueries

Learn:

- Scalar subquery
- Multi-row subquery
- Correlated subquery
- Nested subquery

Practice:

- Employees above average salary.
- Customers above average spending.
- Products above average price.
- Customers with orders.
- Products never sold.

Day 13 — EXISTS / NOT EXISTS

Master:

EXISTS
NOT EXISTS

Practice:

- Customers with orders.
- Customers without orders.
- Products sold.
- Products never sold.
- Employees assigned to departments.
- Departments without employees.

Important interview topic:

> `EXISTS` vs `IN` vs `JOIN`

Day 14 — Week 2 Project

Sales Analysis System

Create queries for:

Total Revenue
Revenue by Month
Revenue by Customer
Revenue by Product
Revenue by Category
Top 10 Customers
Top 10 Products
Customers Without Orders
Products Without Sales
Average Order Value

You should now be comfortable with **basic/intermediate SQL**.

Week 3 — Advanced SQL

Day 15 — CTE

Learn:

WITH CTE AS (...)

Topics:

- Simple CTE
- Multiple CTEs
- CTE + JOIN
- CTE + aggregation
- Recursive CTE

Practice:

- Customer sales CTE.
- Department salary CTE.
- Top customers.
- Monthly sales.
- Duplicate detection.

Day 16 — ROW_NUMBER

Master:

ROW_NUMBER() OVER(...)

Practice:

61. Latest order per customer.
62. First order per customer.
63. Top 3 products per category.
64. Top 3 employees per department.
65. Remove duplicates.
66. Latest customer record.
67. Latest claim per patient.

Day 17 — RANK & DENSE_RANK

Understand:

ROW_NUMBER
RANK
DENSE_RANK

Example:

Salary

100000
100000
90000
80000

Understand how each function assigns rankings.

Practice:

- Top salaries.
- Second-highest salary.
- Third-highest salary.
- Ranking products.
- Ranking customers.

Day 18 — LAG & LEAD

Learn:

LAG()
LEAD()

Practice:

68. Previous order amount.
69. Next order amount.
70. Difference between current and previous sale.
71. Customer purchase interval.
72. Month-over-month sales.
73. Previous claim amount.
74. Previous employee salary.

Day 19 — Window Aggregations

Learn:

SUM() OVER()
AVG() OVER()
COUNT() OVER()
MIN() OVER()
MAX() OVER()

Practice:

75. Running sales total.
76. Running customer spending.
77. Cumulative quantity.
78. Moving average.
79. Customer percentage contribution.
80. Department salary percentage.

Day 20 — Advanced Analytical SQL

Learn:

- FIRST_VALUE
- LAST_VALUE
- NTILE
- Percentiles

- Running totals
- Moving averages
- Cumulative percentages

Practice:

- First purchase.
- Last purchase.
- Customer percentile.
- Top 10%.
- Rolling 7-day sales.
- Rolling 30-day sales.

Day 21 — Advanced SQL Project

Customer Analytics

Create a report containing:

Customer
FirstOrderDate
LastOrderDate
TotalOrders
TotalSales
AverageOrderValue
PreviousOrderAmount
NextOrderAmount
RunningSales
CustomerRank
CustomerSegment

This should combine:

- CTE
- JOIN
- CASE
- GROUP BY
- Window functions
- LAG
- LEAD

Week 4 — Data Engineering SQL

Day 22 — Deduplication

Learn:

ROW_NUMBER
PARTITION BY
ORDER BY

Solve:

81. Find duplicate customers.
82. Find duplicate transactions.
83. Keep latest record.
84. Keep oldest record.
85. Remove duplicate records.
86. Identify duplicate claims.

This is **very common in Data Engineer interviews**.

Day 23 — Incremental Loading

Learn:

Full Load
Incremental Load
Watermark
LastModifiedDate
High-water mark

Example:

WHERE ModifiedDate > @LastLoadDate

Practice:

87. Extract today's records.
88. Extract records modified since last load.
89. Identify new customers.
90. Identify updated customers.
91. Create incremental load query.

Day 24 — SCD Type 1 & Type 2

Learn:

Type 1

Overwrite:

Old City → New City

Type 2

Preserve history:

Customer
Old Record → Expired
New Record → Current

Practice:

92. Detect changed customers.
93. Identify new customers.
94. Implement Type 1.
95. Implement Type 2.
96. Create `IsCurrent`.
97. Create StartDate/EndDate.

Day 25 — MERGE / UPSERT

Learn:

INSERT
UPDATE
UPSERT
MERGE

Practice:

- Insert new customers.
- Update existing customers.
- Synchronize staging and target.
- Implement SCD logic.
- Compare source and target.

Interview:

How would you implement an incremental pipeline using SQL?

Day 26 — Transactions

Learn:

BEGIN TRANSACTION
COMMIT
ROLLBACK
SAVE TRANSACTION

And:

ACID
Isolation Levels
Locks
Blocking
Deadlocks

Practice:

98. Bank transfer transaction.
99. Order + payment transaction.
100. Rollback failed transaction.

Day 27 — SQL Performance

This is critical for a senior Data Engineer.

Learn:

Indexes

- Clustered
- Nonclustered
- Composite
- Covering
- Included columns
- Filtered indexes

Execution Plans

Understand:

Index Seek
Index Scan
Table Scan
Key Lookup
Hash Match
Nested Loop
Merge Join
Sort

Interview questions:

- Why is a query slow?
- What is an index seek?
- Index scan vs seek?
- Clustered vs nonclustered?
- What is a covering index?
- What is SARGability?

Day 28 — Data Warehouse SQL

Learn:

Fact Table
Dimension Table
Star Schema
Snowflake Schema
Surrogate Key
Natural Key
Degenerate Dimension
Conformed Dimension
Date Dimension

Build:

DimCustomer
DimProduct
DimDate
DimStore

FactSales

Then write SQL queries against the warehouse.

Day 29 — Azure/Fabric SQL

Connect SQL knowledge to your Data Engineer stack.

Study:

Azure SQL

- Azure SQL Database
- SQL Managed Instance

Azure Synapse

- Dedicated SQL pool
- Serverless SQL
- External tables
- CETAS
- COPY INTO

Microsoft Fabric

- Fabric Warehouse
- Lakehouse
- SQL Analytics Endpoint
- OneLake
- Delta tables

Data pipeline

SQL Server
↓
ADF
↓
ADLS Gen2
↓
Fabric / Databricks
↓
Delta
↓
Warehouse
↓
Power BI

Interview question:

> Explain how you would implement incremental ingestion from Azure SQL into a Fabric Lakehouse.

Day 30 — Final SQL Interview Day

Part 1 — SQL Assessment

Take a 2-hour test covering:

Fundamentals

- SELECT
- WHERE
- GROUP BY
- HAVING
- JOIN

Intermediate

- CASE
- Subqueries
- CTE
- EXISTS
- Date functions

Advanced

- Window functions
- LAG/LEAD
- Ranking
- Deduplication
- Running totals

Data Engineering

- Incremental load
- SCD Type 2
- MERGE
- CDC
- Data warehouse

Your 100-Question Distribution

Topic	Questions
SELECT / Filtering	10
Sorting / DISTINCT / TOP	10
Aggregation	10
GROUP BY / HAVING	10
JOINs	10
CASE / NULL / Functions	10
Subqueries / EXISTS / CTE	10
Window Functions	15
Deduplication / Analytics	5
ETL / Incremental / SCD	5
Transactions / Performance	5
Total	**100**

30-Day Interview Preparation

Every day, don't just solve queries. Use this process:

Step 1 — Understand

Read the requirement.

Step 2 — Identify

Ask:

What tables?
What relationship?
What columns?
What filtering?
What aggregation?
What ranking?

Step 3 — Write

Write the query yourself.

Step 4 — Test

Run it against the dataset.

Step 5 — Optimize

Ask:

> Can I make this query more efficient?

Step 6 — Explain

Explain the query verbally as if you're in an interview.

Most Important 25 SQL Interview Questions

You should be able to answer these without hesitation:

1. WHERE vs HAVING?
2. INNER JOIN vs LEFT JOIN?
3. UNION vs UNION ALL?
4. DELETE vs TRUNCATE vs DROP?
5. Primary key vs unique key?
6. Clustered vs nonclustered index?
7. CTE vs temporary table?
8. CTE vs subquery?
9. RANK vs DENSE_RANK vs ROW_NUMBER?
10. EXISTS vs IN?
11. What is a correlated subquery?
12. How do you find duplicates?
13. How do you remove duplicates?
14. How do you find the second-highest salary?
15. How do you find top 3 employees per department?
16. How do you find customers with no orders?
17. How do you calculate a running total?
18. How do you calculate month-over-month growth?
19. How do you find the latest record per customer?
20. How do you implement SCD Type 2?
21. What is an incremental load?
22. What is CDC?
23. How do you optimize a slow SQL query?

24. Explain an execution plan.
25. Explain your SQL solution from a real Data Engineering project.

The Final Skill Target

By the end of these 30 days, you should be able to take a requirement such as:

> "Find the top 3 customers in each state based on total spending during the last 12 months, compare their spending with the previous month, remove duplicate customer transactions, and return only active customers."

and independently determine:

JOIN
↓
WHERE
↓
GROUP BY
↓
CTE
↓
ROW_NUMBER()
↓
LAG()
↓
CASE
↓
Final filtering

That is the level of SQL I would target for a **senior Azure/Fabric Data Engineer interview**.